

```
Create DATABASE HospitalDB;  
GO
```

```
USE HospitalDB;  
GO
```

```
-- Doctors
```

```
CREATE TABLE Doctors (  
  doctor_id VARCHAR(10) PRIMARY KEY,  
  doctor_name VARCHAR(100) NOT NULL,  
  hospital_unit VARCHAR(100) NOT NULL  
);
```

```
--Patients
```

```
Create table Patients (  
  patient_id INT IDENTITY(1,1) PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  doctor_id VARCHAR(10) NOT NULL,  
  diagnosis VARCHAR(100) NOT NULL,  
  contact_number VARCHAR(15),  
  FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
);
```

```
-- Appointments
```

```
CREATE TABLE Appointments (  
  appointment_id INT IDENTITY(1,1) PRIMARY KEY,  
  patient_id INT NOT NULL,  
  doctor_id VARCHAR(10) NOT NULL,  
  appointment_date DATE NOT NULL,  
  CONSTRAINT FK_Appointments_Patients FOREIGN KEY (patient_id) REFERENCES Patients  
    (patient_id),  
  CONSTRAINT FK_Appointments_Doctors FOREIGN KEY (doctor_id) REFERENCES Doctors  
    (doctor_id)  
);
```

```
-- Treatments
```

```
CREATE TABLE Treatments (  
  treatment_id INT IDENTITY(1,1) PRIMARY KEY,  
  diagnosis VARCHAR(100),  
  prescription VARCHAR(100),  
  cost DECIMAL(10,2),  
  appointment_ID INT NOT NULL,  
  FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id)  
);
```

```
INSERT INTO Doctors VALUES
```

```
( 'D101', 'Naledi Khumalo', 'Pediatrics' ),
( 'D102', 'Thabo Maseko', 'Cardiology' ),
( 'D103', 'Aisha Patel', 'Dermatology' ),
( 'D104', 'Michael Smith', 'Orthopedics' ),
( 'D105', 'Lerato Dube', 'General Surgery' );
```

```
INSERT INTO Patients (name, doctor_id, diagnosis, contact_number) VALUES
( 'Sipho Ndlovu', 'D101', 'Asthma', '0712345678' ),
( 'Nomsa Mthembu', 'D102', 'Hypertension', '0723456789' ),
( 'David Johnson', 'D103', 'Eczema', '0734567890' ),
( 'Zanele Nkosi', 'D104', 'Fracture', '0745678901' ),
( 'Peter Adams', 'D105', 'Appendicitis', '0756789012' ),
( 'Ayanda Molefe', 'D101', 'Bronchitis', '0767890123' ),
( 'Sarah Jacobs', 'D102', 'Diabetes', '0778901234' ),
( 'John Williams', 'D103', 'Psoriasis', '0789012345' ),
( 'Kabelo Moeketsi', 'D104', 'Sprain', '0790123456' ),
( 'Emily Brown', 'D105', 'Gallstones', '0801234567' );
```

```
INSERT INTO Appointments (patient_id, doctor_id, appointment_date) VALUES
(1, 'D101', '2026-07-01' ),
(2, 'D102', '2026-07-02' ),
(3, 'D103', '2026-07-03' ),
(4, 'D104', '2026-07-04' ),
(5, 'D105', '2026-07-05' ),
(6, 'D101', '2026-07-06' ),
(7, 'D102', '2026-07-07' ),
(8, 'D103', '2026-07-08' ),
(9, 'D104', '2026-07-09' ),
(10, 'D105', '2026-07-10' );
```

```
INSERT INTO Treatments (diagnosis, prescription, cost, appointment_id) VALUES
( 'Asthma', 'Inhaler', 250.00, 1 ),
( 'Hypertension', 'Amlodipine', 180.00, 2 ),
( 'Eczema', 'Topical cream', 120.00, 3 ),
( 'Fracture', 'Cast and painkillers', 500.00, 4 ),
( 'Appendicitis', 'Surgery', 1500.00, 5 ),
( 'Bronchitis', 'Antibiotics', 300.00, 6 ),
( 'Diabetes', 'Insulin', 400.00, 7 ),
( 'Psoriasis', 'Steroid cream', 220.00, 8 ),
( 'Sprain', 'Physiotherapy', 350.00, 9 ),
( 'Gallstones', 'Surgery', 2000.00, 10 );
```

--Validation queries

```
SELECT * FROM Patients;
```

-- Join patients with doctors and appointments

```
SELECT p.name, d.doctor_name, a.appointment_date
FROM Patients p
JOIN Appointments a ON p.patient_id = a.patient_id
```

```
JOIN Doctors d ON a.doctor_id = d.doctor_id;
```

```
-- Show treatment with costs
```

```
SELECT t.diagnosis, t.prescription, t.cost, a.appointment_date  
FROM Treatments t  
JOIN Appointments a ON t.appointment_id = a.appointment_id;
```

```
--Top 3 most expensive treatments
```

```
SELECT TOP 3 prescription, cost  
FROM Treatments  
ORDER BY cost DESC;
```

```
-- Number of patients per doctor
```

```
SELECT d.doctor_name, COUNT(p.patient_id) AS patient_count  
FROM Doctors d  
LEFT JOIN Patients p ON d.doctor_id = p.doctor_id  
GROUP BY d.doctor_name;
```

```
SELECT diagnosis, AVG(cost) AS avg_cost  
FROM Treatments  
GROUP BY diagnosis;
```

```
--Views
```

```
CREATE VIEW PatientTreatmentSummary AS  
SELECT p.name AS patient_name,  
       d.doctor_name,  
       a.appointment_date,  
       t.diagnosis,  
       t.prescription,  
       t.cost  
FROM Patients p  
JOIN Appointments a ON p.patient_id = a.patient_id  
JOIN Doctors d ON a.doctor_id = d.doctor_id  
JOIN Treatments t ON a.appointment_id = t.appointment_id;
```

```
--Stored Procedures
```

```
CREATE PROCEDURE GetPatientHistory @patient_id INT  
AS  
BEGIN  
    SELECT p.name, d.doctor_name, a.appointment_date, t.diagnosis, t.prescription,   
           t.cost  
    FROM Patients p  
    JOIN Appointments a ON p.patient_id = a.patient_id  
    JOIN Doctors d ON a.doctor_id = d.doctor_id  
    JOIN Treatments t ON a.appointment_id = t.appointment_id  
    WHERE p.patient_id = @patient_id;  
END;
```

